Understanding Networking Principles

Networking principles are the core ideas and rules that explain *how* and *why* networks work. Devices from different makers can talk to each other because they all follow these same principles.

1. Layered Architecture

Networking tasks are split into **layers,** and each layer does one specific job. A layer only communicates with the layer directly above and below it.

Why it matters:

- You can change one layer (e.g., swap Wi-Fi for Ethernet) without rewriting the rest.
- Troubleshooting is easier because you can isolate which layer has the problem.
- Different vendors can build compatible products.

The OSI model (*7* layers, top to bottom):

Layer	Name	Job	Example
7	Application	Services the user interacts with	HTTP, email
6	Presentation	Formatting, encryption, compression	SSL/TLS, JPEG
5	Session	Starts, manages, ends conversations	Login sessions
4	Transport	Reliable or fast delivery, ports	TCP, UDP
3	Network	Addressing and routing	IP, routers
2	Data Link	Delivery within one network, MAC addresses	Ethernet, switches
1	Physical	Raw bits over a medium	Cables, radio signals

Memory aid, from layer *7* down: "*All People Seem To Need Data Processing*."

2. Encapsulation

When you send data, each layer **wraps it with its own header** (like putting a letter in an envelope, then in a bigger envelope). The receiving device **de-encapsulates** it by removing the wrappers in reverse order.

Layer	Data unit name
Transport	Segment (TCP) / Datagram (UDP)
Network	**Packet** (adds IP addresses)
Data Link	**Frame** (adds MAC addresses)
Physical	**Bits**

3. Packet Switching

Data is broken into small **packets**. Each packet travels independently, possibly by a different route, and the destination reassembles them in order.

- **Advantages:** efficient use of bandwidth, and the network keeps working if one path fails.
- **Compare with circuit switching:** the old telephone system reserved one dedicated path for the whole call, which wasted capacity when nobody was talking.

4. Addressing and Delivery Types

Every device needs an address so data reaches the right place.

- **Unicast:** one sender to one receiver (loading a web page).
- **Broadcast:** one sender to all devices on the network (DHCP discovery, ARP).
- **Multicast:** one sender to a selected group (IPTV, video conferencing).

5. Standards and Protocols

A **protocol** is an agreed set of rules for communication, and a **standard** is a formally published specification.

- **IEEE:** Ethernet (802.3) and Wi-Fi (802.11)
- **IETF:** TCP/IP, HTTP, DNS
- **TIA/EIA:** cabling standards

6. Reliability and Performance Controls

- **Error detection:** checksums and CRC reveal corrupted data.
- **Flow control:** stops a fast sender from overwhelming a slow receiver.
- **Congestion control:** slows traffic when the network is overloaded.
- **Duplex modes:** *half-duplex* sends or receives one at a time (walkie-talkie), and *full-duplex* does both at once (phone call). Modern switches use full-duplex.

7. Collision and Broadcast Domains

- **Collision domain:** an area where two devices sending at once would interfere. A hub puts all devices in one domain, while a **switch** gives each port its own.
- **Broadcast domain:** the area a broadcast message reaches. A **router** (or VLAN) separates broadcast domains.

8. Security Principles: The CIA Triad

Principle	**Meaning**	**Example tool**
Confidentiality	Only authorised people can read the data	Encryption
Integrity	Data has not been changed	Hashing

Identifying Common Network Hardware

1. Network Interface Card (NIC)

The hardware inside a device that connects it to the network, either wired (Ethernet) or wireless (Wi-Fi). Every NIC has a unique **MAC address**. NICs get their own full topic (Topic 4), so this is only a quick mention.

2. Hub (obsolete)

A hub is a simple device with several ports. It receives data on one port and **repeats it out of all other ports**, whether or not every device needs it.

- **Layer:** 1 (Physical)
- **Problems:** all devices share one collision domain, bandwidth is shared, and any device can see everyone's traffic (insecure).
- **Status:** replaced by switches. You may still see it in exam questions.

3. Switch

A switch connects devices within a LAN. It builds a **MAC address table** by learning which device is on which port, then sends each frame **only to the correct port**.

- **Layer:** 2 (Data Link). *Layer 3 switches* can also route.
- **Advantages:** each port is its own collision domain, it supports full-duplex, and it gives better speed and privacy than a hub.
- **Unmanaged:** plug-and-play, with no configuration (home and small office).
- **Managed:** configurable, with VLANs, port security, monitoring, and QoS (businesses).
- **PoE (Power over Ethernet) switch:** also powers devices such as IP phones, cameras, and access points through the network cable.

4. Router

A router connects **different networks** (for example, your LAN to the internet) and chooses the best path for packets using **IP addresses** and a routing table.

- **Layer:** 3 (Network)
- **Separates broadcast domains**, so broadcasts don't cross it.
- **Home router:** usually an all-in-one box with a router, a 4-port switch, a firewall, NAT, DHCP, and a Wi-Fi access point.
- **Enterprise router:** handles WAN links, routing protocols (OSPF, BGP), and VPNs.

5. Wireless Access Point (WAP/AP)

An access point provides **Wi-Fi** and connects wireless devices to the wired network.

- **Layer:** 2
- Larger buildings use several APs with the same network name so users can roam between them.
- Managed centrally by a **wireless LAN controller** in larger organisations.
- **Difference from a wireless router:** a wireless router *includes* routing, while an AP only adds wireless access to an existing network.

6. Modem

A modem **modulates and demodulates** signals so your network can use the ISP's line.

Connection	Device
DSL (phone line)	DSL modem
Cable TV line	Cable modem
Fibre	ONT (Optical Network Terminal)

Many ISPs supply a combined modem and router.

7. Firewall

A firewall filters traffic using security rules and decides what to allow or block.

- **Hardware firewall:** a dedicated appliance at the network edge.
- **Software firewall:** runs on a PC or server (e.g., Windows Defender Firewall).
- **Types:** packet filtering (Layers 3-4), stateful inspection, and next-generation firewalls (NGFW) that inspect applications (up to Layer 7).

8. Other Common Devices

Device	Purpose
Repeater / Extender	Boosts a weak signal to extend range
Bridge	Joins two network segments and filters by MAC (switches are multi-port bridges)
Patch panel	Organises cable terminations in a comms room, so devices connect through short patch cables
NAS	Network Attached Storage, shared file storage on the network
Load balancer	Spreads traffic across several servers
Gateway	Connects networks that use different protocols
Proxy server	Sits between users and the internet, caching and filtering requests

Hub vs. Switch vs. Router

	Hub	Switch	Router
OSI layer	1	2	3
Uses	Nothing (repeats bits)	MAC addresses	IP addresses
Sends data to	All ports	One specific port	The best path to another network
Collision domains	One for all	One per port	One per interface
Broadcast domains	One	One (unless VLANs)	Separates them
Connects	Devices in a LAN	Devices in a LAN	Different networks

Typical Home Network Layout

Internet → **Modem/ONT** → **Router** (with firewall, NAT, DHCP) → **Switch** and **Access Point** → PCs, printers, phones, and smart devices.

Key Points to Remember

- **Hub** repeats to everyone, **switch** forwards by MAC, **router** forwards by IP.
- A **router** links different networks, and a **switch** links devices within one.
- An **access point** adds Wi-Fi, and a **modem** connects to the ISP.
- A **firewall** filters traffic by rules.
- Managed switches add VLANs and monitoring, and PoE powers devices over the cable.

Network Interface Cards (NICs)

1. What is a NIC?

A **Network Interface Card** is the hardware that lets a computer send and receive data on a network. It converts the computer's digital data into signals suited to the medium (electrical for copper, light for fibre, radio for Wi-Fi) and converts incoming signals back into data.

- **OSI layers:** 1 (Physical) and 2 (Data Link)
- **Other names:** network adapter, LAN card, Ethernet card, NIC
- **Connector:** RJ-45 for copper Ethernet, SFP/LC for fibre, or an antenna for Wi-Fi

2. The MAC Address

Every NIC has a unique **48-bit MAC (Media Access Control) address**, written in hexadecimal, such as A4-5E-60-C2-1B-7F.

- **First half (OUI):** identifies the manufacturer.

- **Second half:** a unique number assigned to that card.
- Used for delivery **within a local network** (switches use it to forward frames).
- Burned in at the factory, though it can be changed in software ("MAC spoofing").

3. Types of NIC

Type	Description	Typical use
Onboard (integrated)	Built into the motherboard	Most desktops and laptops
PCIe card	Plugs into an expansion slot	Upgrades, multi-port or 10 Gbps cards
USB adapter	Plugs into a USB port	Laptops with no Ethernet port, quick fixes
Wireless (Wi-Fi) NIC	Radio-based, with antennas	Laptops, phones, home PCs
Server NIC	Multi-port, with advanced offload features	Servers and virtualisation hosts
Fibre NIC	Uses SFP/SFP+ modules	Data centres, high-speed links

Older PCs used **PCI** cards, which have been replaced by **PCIe**.

4. NIC Speeds

Speed	Name	Typical use
10 Mbps	Ethernet	Obsolete
100 Mbps	Fast Ethernet	Old devices
1 Gbps	Gigabit Ethernet	Standard today
2.5 / 5 / 10 Gbps	Multi-gig	Workstations, NAS, servers
25-100 Gbps		Data centres

A NIC only runs as fast as the **slowest part of the path**: the NIC, cable, and switch port must all support the speed.

5. Duplex and Auto-Negotiation

- **Half-duplex:** send *or* receive at one time.
- **Full-duplex:** send *and* receive at the same time (standard today).
- **Auto-negotiation:** the NIC and switch automatically agree on the best speed and duplex.
- A **duplex mismatch** (one side full, the other half) causes slow speeds and many errors.

6. Wired vs. Wireless NICs

	Wired	**Wireless**
Speed	Higher and consistent	Variable
Latency	Low	Higher
Reliability	Very high	Affected by interference and distance
Security	Physical access needed	Needs encryption (WPA2/WPA3)
Best for	Servers, workstations, gaming	Mobility and convenience

Wi-Fi standards:

Standard	**Name**	**Bands**
802.11n	Wi-Fi 4	2.4 / 5 GHz
802.11ac	Wi-Fi 5	5 GHz
802.11ax	Wi-Fi 6 / 6E	2.4 / 5 / 6 GHz
802.11be	Wi-Fi 7	2.4 / 5 / 6 GHz

7. Important NIC Features

- **Wake-on-LAN (WoL):** powers on a PC remotely over the network.
- **PXE boot:** boots an operating system from the network (used for thin clients and mass deployments).
- **NIC teaming (bonding):** combines several NICs for more speed or for redundancy.
- **TCP offload:** moves network processing from the CPU to the NIC.
- **SR-IOV:** lets virtual machines share a NIC efficiently.
- **VLAN tagging:** supports VLANs directly on the card.
- **Promiscuous mode:** the NIC accepts all frames it sees, not only its own (used by packet sniffers such as Wireshark).
- **Link LEDs:** a steady light shows a connection, and a blinking light shows activity.

8. Installing and Configuring a NIC

1. **Install the hardware:** seat a PCIe card in a slot, or plug in a USB adapter.
2. **Install the driver:** Windows often does this automatically, or you can download it from the manufacturer.
3. **Connect the cable** to a switch or router.
4. **Configure IP settings:** automatically through DHCP, or manually (IP, subnet mask, gateway, DNS).
5. **Verify** the connection is working.

9. Troubleshooting a NIC

Check	How
Physical	Is the cable plugged in? Are the link LEDs on?
Driver	Check Device Manager for a warning icon, then update or reinstall the driver
Enabled	Make sure the adapter is not disabled in settings
IP settings	ipconfig /all (Windows) or ip a (Linux)
Own stack	ping 127.0.0.1 (loopback) tests the TCP/IP software on the machine
Gateway	ping the default gateway to test the local network
Outside	ping 8.8.8.8 tests internet access, and ping google.com tests DNS

Common problems:

- **169.254.x.x address (APIPA):** the PC could not reach a DHCP server.
- **No link light:** a bad cable, a bad port, or a disabled NIC.
- **Slow speed:** a duplex mismatch, a poor cable, or a Cat5 cable limiting the speed.

Key Points to Remember

- A **NIC** connects a device to the network and works at **Layers 1 and 2**.
- Every NIC has a unique **48-bit MAC address**.
- Types: onboard, PCIe, USB, wireless, server, and fibre.
- Speed depends on the NIC, cable, and switch port together.
- **WoL, PXE, teaming, and SR-IOV** are key advanced features.
- A **169.254.x.x** address means DHCP failed.